

MASTER PROGRAM

CYBERSECURITY MANAGEMENT AND DATA SOVEREIGNTY

(ADVANCED CYBER SECURITY)

Description of Modules in the Curriculum

Law, Compliance, Governance, Policy, and Ethics

This module focuses on equipping students with an in-depth understanding of the legal, ethical, and governance frameworks that shape cybersecurity practices. It provides comprehensive insights into data protection laws, governance structures, and the intersection of legal and ethical standards in organizational cybersecurity strategies.

The learning outcomes of the module are:

- Integrates Ethical Cybersecurity Practices: Integrates and promotes ethical considerations in decision-making and conduct across all levels of the organization, enhancing organizational integrity through privacy, confidentiality, and accountability.
- Champion Diversity and Inclusion: Creates and implements diversity and inclusion strategies within cybersecurity roles and policies to foster innovation, resilience, and organizational effectiveness.
- Communicate Complex Concepts: Communicate complex legal, ethical, and policy issues to diverse stakeholders to facilitate informed decision-making and ensure compliance.
- Analyzes Legal and Ethical Frameworks: Analyze and critically evaluate legal frameworks and ethical standards in cybersecurity, including the overlap and differences between cybersecurity, data protection, and privacy.
- Interprets and Apply Cybersecurity Laws and Regulations: Interpret and apply legal requirements to protect information assets, ensuring compliance with international regulations and internal governance tools such as policies or standards.
- Compares and Evaluates Policies: Compares and assess policies that address ethical, legal, and practical aspects of cybersecurity operations while promoting organizational integrity.

Module delivery: Giorgio Pedrazzi <giorgio.pedrazzi@unibs.it> (UNIBS), Giorgiana Vlăsceanu <giorgiana.vlasceanu@upb.ro> (UPB)

Cybersecurity Education & Training Delivery I

The module aims to prepare students to identify weaknesses, raise awareness and develop training programs in the realm of cybersecurity education. It presents an array of technical tools and modern methodology to teach cybersecurity related content while combining it with fundamental pedagogical principles.

The learning outcomes of the module are:

- Understands different materials, delivery methods and infrastructure components used in cybersecurity education
- Maps trainee groups requirements with specific types of materials and delivery methods

- Lists the most important stages that marked the development of the field
- Assesses individual and group characteristics with respect to cybersecurity knowledge and skills
- Provides personalized support to trainees

Module delivery: Lisa Moravek <moraveklisa@vut.cz> (BUT), Laura Ruse <laura.ruse@upb.ro> (UPB)

Risk Management of Cyber-Physical Systems

This module aims to equip participants with a comprehensive understanding of risk assessment, management, and mitigation strategies tailored to complex cyber-physical environments. It provides a solid foundation in system safety engineering, probabilistic and socio-technical risk analysis, and the application of advanced risk modeling techniques.

The learning outcomes of the module are:

- Apply risk engineering methods such as FMECA, FTA, ETA, and PRA to assess and analyze vulnerabilities, failure modes, and cascading risks within cyber-physical systems.
- Conduct comprehensive risk assessments of socio-technical systems, including business impact analysis, threat modeling, and scenario planning.
- Develop and implement risk mitigation strategies—such as recovery plans, organizational resilience measures, and security controls—aligned with industry best practices and legal requirements.
- Utilize maturity models (e.g., CMMs, DevSecOps, SBOM) and tools to evaluate and enhance an organization's cyber risk posture.
- Design continuous risk assessment frameworks and integrate them into organizational governance, security policies, and cyber resilience strategies.

Module delivery: Paolo Trucco <paolo.trucco@polimi.it> (POLIMI), Alexandru Văduva <jan.vaduva@upb.ro> (UPB)

Cybersecurity in Industry - Security of OT and Cyber-Physical Systems

This module aims to equip learners with comprehensive knowledge and practical skills in OT (Operational Technology) security, highlighting the differences and overlaps with IT (Information Technology) security. The focus will be on understanding key principles, risk modelling, and the legal and regulatory landscape.

The learning outcomes of the module are:

- Evaluate the principles and challenges of OT security, differentiate them from IT security, and align security strategies with industry standards and best practices.
- Critically assess and compare the OT threat landscape with IT, including analysing specific tactics, techniques, and procedures used in OT-focused cyber-attacks.
- Examine and evaluate recent case studies of cyber incidents in OT environments, drawing lessons on risk assessment and security countermeasures.
- Analyse and apply relevant laws, regulations, and standards to develop robust OT security frameworks, incorporating the latest technological advancements such as AI, 5G, and IIoT.
- Develop comprehensive risk models and security measures tailored to OT environments, ensuring effective governance and continuous risk assessment.



- Assess the impact of human factors on OT security, identifying key vulnerabilities and synthesising remediation measures to strengthen the overall security posture.

Module delivery: Enrico Frumento <enrico.frumento@cefriel.com> (Cefriel), Alexandru Văduva <jan.vaduva@upb.ro> (UPB)

Cybersecurity Economics & Supply Chain

This module aims to equip students with comprehensive knowledge of the economic aspects of information security and supply chain cybersecurity. It covers key concepts such as the direct and indirect costs of cyber incidents, the economic impact of various cyber threats on organizations, and the strategic role of cybersecurity in business planning.

The learning outcomes of the module are:

- Apply economic analysis techniques to assess the financial impact of cyber incidents on organizations and supply chains.
- Evaluate different cybersecurity investment scenarios to determine optimal budget allocation for security controls and risk mitigation.
- Analyze vulnerabilities and risks in supply chain cybersecurity, including third-party risks, and recommend mitigation strategies based on economic impacts.
- Utilize frameworks and standards to develop cost-effective cybersecurity strategies aligned with organizational goals.
- Interpret emerging cybersecurity trends and technologies to anticipate their economic effects on organizational investments and supply chain resilience.

Module delivery: Marius Laurinaitis <laurinaitis@mrui.eu> (MRU), Florin Anton <florin.anton@upb.ro> (UPB)

Cybersecurity Auditor

This module aims at providing students with skills and understanding to build an annual audit programme and to develop an audit plan for specific audit assignments. Also, students would be able to conduct auditing assignments and produce the resulting report.

The learning outcomes of the module are:

- Implement audit activities as part of the assurance process, including scoping, the selection of suitable criteria, the audit process and the audit reporting.
- Plan, develop and conduct a full audit assignment on Cybersecurity activities with the aim of informing stakeholders on the maturity, effectiveness and outcome of audit activities to align with business requirements and good auditing and business practices.
- Create the audit plan, adjust the scoping of the assignment to business and technical needs or stakeholders request
- Develop a scoping of the assignment; select a suitable criteria; align with usual audit practices including the Certificate in IT auditing.
- Develop an audit fieldwork and produce a report with findings and recommendations.

Module delivery: Georges Ataya <ga@atayapartners.com> (Solvay), Costin Carabaș <costin.carabas@upb.ro> (UPB)

Enterprise Architecture, Infrastructure Design and Cloud Computing

The master class aims to provide the knowledge and skills related to modern enterprise security and infrastructure principles, including threat landscapes, network security architecture, critical security controls, and cloud security strategies.

The learning outcomes of the module are:

- Apply techniques and tools for vulnerability assessment
- Configure and optimize security controls such as IDS/IPS, firewalls, and encryption to prevent and detect cyberattacks.
- Design and implement security solutions for applications and cloud infrastructure
- Develop security policies and procedures for various scenarios
- Configure and use dashboards and monitoring tools to track security status and respond effectively to security events.

Module delivery: Gillian O'Carroll <gillian.ocarroll@mtu.ie> (MTU), Ciprian Dobre <ciprian.dobre@upb.ro> (UPB)

Cybersecurity Culture, Strategy & Leadership

This module aims to equip participants with a comprehensive understanding of the cybersecurity landscape, including major threats, vulnerabilities, controls, and regulatory frameworks. It emphasizes the importance of governance, risk management, incident response, and maturity monitoring in creating an effective cybersecurity strategy.

The learning outcomes of the module are:

- Analyze cybersecurity incidents and extract lessons to inform organization-specific security strategies.
- Apply frameworks and controls to evaluate organizational security posture and identify gaps.
- Develop a cybersecurity risk management plan aligned with business objectives and regulatory requirements.
- Design a comprehensive cybersecurity roadmap covering the phases of protection, detection, and response.
- Create dashboards and reports to monitor cybersecurity maturity and effectiveness, integrating technical and managerial indicators.

Module delivery: Georges Ataya <ga@atayapartners.com> (Solvay), Răzvan Rughiniș <razvan.rughinis@upb.ro> (UPB)

Communication Design for Cybersecurity

The overall objectives of this module are to equip participants with the knowledge, skills, and responsibilities needed to develop and implement effective cybersecurity communication strategies, focusing on creating clear, persuasive, and ethical messages tailored to diverse audiences and contexts.

The learning outcomes of the module are:

- Apply communication frameworks to develop targeted messages for different stakeholders, ensuring clarity and effectiveness.
- Analyze cybersecurity communication scenarios to identify vulnerabilities, malicious tactics, and areas for improvement.
- Design compelling and ethically responsible communication strategies, including crisis messaging and stakeholder engagement.
- Create and deliver persuasive presentations of cybersecurity strategies, integrating data visualization and storytelling techniques.
- Utilize tools such as stakeholder mapping, message testing, and rapid prototyping to enhance communication effectiveness.

Module delivery: Julia von Thienen <julia.vonthienen@german-uds.de> (UDS), Florin Pop <florin.pop@upb.ro> (UPB)

AI & Emerging Topics in Cybersecurity

The module delves into AI and its impact on cybersecurity, highlighting its offensive and defensive applications. Using both lectures and flipped classroom sessions, students will learn key AI concepts and their applications in today's Security Operations Centers (SOCs).

The learning outcomes of the module are:

- Identify security issues associated with AI models, such as vulnerabilities, weaknesses, and adversarial threats.
- Performs research and specify emerging trends in AI and cybersecurity.
- Designs and develops AI-driven solutions for real-world cybersecurity challenges, demonstrating proficiency in both offensive and defensive applications.
- Critically evaluates the strengths, and limitations of AI applications in cyberdefense.
- Identifies the potential misuse of AI by adversaries to automate and enhance TTPs

Module delivery: Pejman Najafi <pejman.najafi@german-uds.de> (UDS), Radu Marin <radu.marin@upb.ro> (UPB)

CISO and Crisis Communication

This module aims to build the essential skill required in crisis situations and as regular communication of the CISO to stakeholders. Crisis communication is required for three specific focuses that are: incident resolution, compliance notification, improving reputation.

The learning outcomes of the module are:

- Recognize crisis communication for the three purposes of containing the incident/crisis, of regulatory notification, and of preserving reputation.
- Recognize regular awareness activities as a n essential control for improving the preparedness of the Human Factor.
- Master the various actions related to planning, developing and conducting communication actions.
- Create the communication plan including the design, the timeline, the identification of target audience, the definition of the communication channel, the identification of the message, the evaluation of the impact, and the improvement for a future cycle.
- Interpretation of causal relationships.



- Identify solutions and proposes resolution plans/projects

Module delivery: Georges Ataya <ga@atayapartners.com> (Solvay), Cătălin Goșman <catalin.gosman@upb.ro> (UPB)

Security Operations

This module aims to provide participants with a comprehensive understanding of Security Operations Centers (SOCs), their functions, tools, and technologies; network and endpoint security strategies; and incident detection, investigation, and response techniques.

The learning outcomes of the module are:

- Develop scripts and configurations for managing and monitoring firewalls, IPS, EDR, and other SOC technologies to identify and prevent threats.
- Apply techniques for collecting, categorizing, and analyzing security data for incident investigation and response.
- Design and automate SOC workflows to improve processes and increase the speed of incident handling.
- Use event correlation and normalization tools to detect suspicious activities and complex cyberattacks.
- Perform advanced malware analysis, reverse engineering, and threat hunting to identify and counteract new attack vectors.

Module delivery: Michel Guilmault <michel.guilmault@cyu.fr> (CYU), Radu Ciobanu <radu.ciobanu@upb.ro> (UPB)

Technological Foundations for CS & Security Controls

The module aims to build a solid foundation for understanding computer systems, managing data securely, developing secure applications, and protecting systems now and in future technological environments.

The learning outcomes of the module are:

- Investigate hardware and software configurations of computing systems to assess their components and security posture.
- Install, uninstall, and configure applications and operating system tools effectively.
- Manage data storage, versioning, and backups using appropriate tools and methodologies.
- Apply secure programming practices, including code review, static/dynamic analysis, and defensive coding techniques.
- Configure network parameters, troubleshoot connectivity issues, and implement basic network security measures.

Module delivery: Razvan Deaconescu <razvan.deaconescu@upb.ro> (UPB), Florin Anton <florin.anton@upb.ro> (UPB)

Business Resilience, Incident Management and Threat Response

This module aims to equip participants with a comprehensive understanding of industry-leading best practices—including NIST, ISO, and other standards—for establishing and enhancing organizational resilience against cyber threats. It fosters the ability to assess and mitigate risks, develop effective incident response strategies, leverage threat intelligence, and foster a culture of continuous improvement.

The learning outcomes of the module are:

- Apply best practices and frameworks to assess organizational risk, response preparedness, and compliance requirements in real-world scenarios.
- Develop and execute incident response plans, incorporating detection, investigation, analysis, and activation procedures tailored to organizational needs.
- Utilize security tools such as SIEM, forensic, and E-discovery solutions to investigate and analyze security incidents from end to end.
- Perform Business Impact Analyses and process mapping to evaluate organizational resilience and operational continuity.
- Design and implement incident response workflows and escalation procedures to improve response effectiveness and minimize disruption.

Module delivery: Mihai Carabaș <mihai.carabas@upb.ro> (UPB)

Automation of Security Tasks and Data Analytics

This module aims to equip participants with the skills and knowledge to develop, implement, and manage automated cybersecurity workflows using Python and related tools. It emphasizes practical skills in a diverse range of activities from automating reconnaissance, to vulnerability management and tool orchestration.

The learning outcomes of the module are:

- Implement Python scripts to perform reconnaissance activities, open-source intelligence gathering, and network scanning using appropriate libraries.
- Develop automated scripts for threat intelligence collection, data analysis, visualization, and log monitoring applying Python data manipulation libraries.
- Create scripts for automating incident detection, response workflows, vulnerability assessments, and exploitation tasks, using system and network libraries.
- Design advanced web scraping and data extraction routines, incorporating browser automation tools, while respecting legal and ethical considerations.
- Build and configure integrated automation solutions using Docker, Ansible, and other orchestration tools to improve security operations.

Module delivery: Sanda Martinčić-Ipšić <smarti@uniri.hr> (UNIRI), Bogdan Mocanu <bogdan_costel.mocanu@upb.ro> (UPB)

Cybersecurity Law & Data Sovereignty

This module aims to provide participants with a comprehensive understanding of the legal frameworks governing cybersecurity and cybercrime, including obligations, liabilities, and tools for ensuring compliance such as certification standards. It also explores the substantive and procedural criminal law applicable to cybercrimes, emphasizing international and European investigation and prosecution tools.

The learning outcomes of the module are:

- Apply legal principles to assess cybersecurity obligations and liabilities in different organizational contexts.
- Analyze case law, legal provisions, and procedural tools to evaluate the best approach for investigation, prosecution, and compliance.
- Develop compliance plans aligned with EU data sovereignty regulations, including measures for processing personal data, data sharing, and electronic evidence handling.
- Utilize international and European procedural tools to support effective investigation and prosecution of cybercrimes.
- Design policies and procedures for handling electronic documents, digital identities, and electronic evidence to ensure legal compliance and data protection.

Module delivery: Petr Dzurenda <dzurenda@vut.cz> (BUT), Giorgia Vlăsceanu <giorgiana.vlasceanu@upb.ro> (UPB)

Research Methods

After taking this module, the students will be able to fully understand the range of research approaches, methodologies and strategies used in research in cybersecurity. The students will have the tools and knowledge by which you can design a research proposal in Cyber Security applying relevant research strategies to collect and test data.

The learning outcomes of the module are:

- Know the range of research approaches, methodologies and strategies used in research within Cyber Security
- Comprehend the tools or knowledge by which they can design a research proposal in Cyber Security applying relevant research strategies to collect and test data;
- Apply evaluation methods in Cyber Security for qualitative and quantitative data;
- Carry out appropriate research analysis in Cyber Security ensuring an ethical research methodology is employed;
- Develop capacity for analysis and synthesis of data instances in Cyber Security
- Develop research skills on exploring real-world issues in Cyber Security

Module delivery: Jan Jürjens <jan.juerjens@isst.fraunhofer.de> (UNI KO), Florin Pop <florin.pop@upb.ro> (UPB)

Cybersecurity Education & Training Delivery II

The module aims to build the required skills for students to be creators of practical cybersecurity contests and use competition-based learning in their educator role. Students will learn how to design, implement, deploy and grade challenges as part of cybersecurity contests (e.g. CTF – Capture the Flag).

The learning outcomes of the module are:

- Recognize common patterns in cybersecurity attack and defense activities that can be replicated and integrated in cybersecurity exercises
- Develop, employ and assess practical cybersecurity exercises, both as single-topic challenges and as vulnerable boxes that feature a complex interconnected topics closer to a real-world setup
- Combine patterns from existing cybersecurity exercises into new customized deployments

- Assess student practical cybersecurity knowledge and skills and revise exercises according to their current level
- Design and set up cyber range-environments and CTF (Capture the Flag)-like contests, including setting up the platform, selecting challenges, grading, providing support
- Summarize and interpret results and feedback of practice activities

Module delivery: Razvan Deaconescu <razvan.deaconescu@upb.ro>

Malware Analysis

This module aims to equip students with the knowledge and skills to analyze and manage the malware analysis process by learning the methods, techniques and tools used. With exercises, videos and laboratories the student will learn the importance and how to use the results for auditing and other cybersecurity processes.

The learning outcomes of the module are:

- Plan and execute the process for malware analysis.
- Analyse the structure and functionality of malware by dissecting code, identifying malicious behaviours, and determining potential damage or security risks.
- Elaborate reports with the results of the malware analysis, for organizational levels
- Evaluate and analyse types of anti-malware solutions according to their performance and know the main malware campaigns to apply preventive measures.
- Manage the malware analysis process during critical time, helping the business resilience plan.
- Develop results reports after malware analysis, for the management taking decision level.

Module delivery: Fidel Paniagua <fidel.paniagua@unir.net> (UNIR), Laura Ruse <laura.ruse@upb.ro> (UPB)

Threat Intelligence

This module aims to introduce the fundamentals of Cyber Threat Intelligence (CTI). The lectures will present the CTI lifecycle, highlight strategic integration and discuss emerging trends in this field. Students will execute data stream identification, assessment and dissemination.

The learning outcomes of the module are:

- Recognize different types of cyber threats and apply analytical techniques to assess their potential impact.
- Gather threat data from open-source and proprietary sources, as well as structure it according to their needs.
- Incorporate threat intelligence into (automated) incident response processes, improving the detection, investigation, and mitigation of attacks.
- Use specialized platforms and tools to analyze threat data and share relevant information.
- Utilize the acquired intelligence to guide proactive threat hunting efforts with the goal of identifying potential compromises and indicators of attack.

Module delivery: Radu Mantu <radu.mantu@upb.ro> (UPB), Bogdan Mocanu <bogdan.mocanu@upb.ro> (UPB)

Ethical Hacking & Penetration Testing

This module will allow the student to understand, analyze and manage the ethical hacking process by learning the concepts, techniques and processes through videos, practical exercises and laboratories. Obtaining the ability to use the results of an audit for management and decision making.

The learning outcomes of the module are:

- Identify threats and attack methodologies during the develop of an ethical Hacking audit.
- Analyse the security posture of systems by identifying vulnerabilities and distinguishing between different types of cyber-attacks.
- Manage the technical and executive reports for taking decisions level.
- Manage ethical hacking audit processes by selecting specific attacks according to the results we need to obtain.
- Plan and manage the ethical hacking process.
- Elaborate results reports after Ethical Hacking process, for the taking decision level

Module delivery: Fidel Paniagua <fidel.paniagua@unir.net> (UNIR), Alexandru Văduva <jan.vaduva@upb.ro> (UPB)

Machine and Deep Learning in Cybersecurity

This module focuses on applying machine learning and deep learning techniques to cybersecurity challenges such as anomaly and malware detection, fraud detection, and spam classification. Students will explore various machine learning algorithms, analyze their performance, and design appropriate models for specific cybersecurity tasks.

The learning outcomes of the module are:

- Compare the advantages and disadvantages of basic machine learning algorithms, especially those related to classification, clustering, and time series analysis.
- Analyse and apply appropriate machine learning methods when solving specific problems such as anomaly or malware detection
- Discuss the possibility of applying machine or deep learning in cybersecurity and explain related problems such as explainability, interpretability, transparency, personal data protection, and legal and ethical challenges.
- Analyse and select deep learning methods that are suitable for the given task in the field of cybersecurity, such as spam detection, and credit card fraud detection.
- Evaluate the performance of the model and, based on that, choose the best machine or deep learning model for a given problem in the field of cybersecurity
- Design and apply a machine or deep learning model for a self-defined problem in the field of cybersecurity

Module delivery: Sanda Martinčić-Ipšić <smarti@uniri.hr> (UNIRI), Dan Novischi <dan_marius.novischi@upb.ro> (UPB)

Digital Forensics, Chain of Custody and eDiscovery

This module aims to equip participants with a comprehensive understanding of digital forensic principles, models, and best practices, emphasizing the importance of evidence handling,



documentation, and legal-ethical obligations. It provides practical knowledge of forensic tools and methodologies across different environments (Windows, Linux, networks, mobile).

The learning outcomes of the module are:

- Apply best practices and standards to identify, acquire, and preserve digital evidence ensuring integrity and admissibility.
- Utilize forensic tools such as drive imaging, password cracking, and log analysis tools to support investigation activities across different environments (Windows, Linux, mobile, network).
- Conduct digital investigations by analyzing data structures, registry information, and network traces, interpreting encryption mechanisms, and recovering deleted files.
- Develop detailed forensic reports, including findings, affidavits, and letters of report, aligned with legal and procedural requirements.
- Implement eDiscovery workflows involving search, keyword matching, deduplication, TAR, and metadata management within eDiscovery platforms like Microsoft Purview or Google Vault.

Module delivery: Răzvan Deaconescu <razvan.deaconescu@upb.ro> (UPB)